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Paper 1: A Quantitative Systems Pharmacology (QSP) Model of Acute Hepatitis B Virus Infection

Bibliographic Information

- **Title:** A Quantitative Systems Pharmacology (QSP) Model of Acute Hepatitis B Virus Infection: Mechanistic Insights and Foundations for Future Extensions
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Quick Take

This paper presents the most comprehensive QSP model of acute HBV infection to date, integrating viral life cycle, innate and adaptive immunity, and cytokine signaling across liver, plasma, and lymph compartments. Researchers in HBV modeling, immunology, and antiviral drug development should read this to understand the mechanistic drivers of infection resolution versus chronicity and to leverage the framework for evaluating novel therapies. The model successfully reproduces clinical biomarker trajectories and identifies key parameters governing viral clearance and immune dynamics.

Executive Summary

This study develops a multiscale QSP model of acute HBV infection that captures viral replication, innate and adaptive immune responses, and cytokine regulation across three anatomical compartments. The model was parameterized using published data and calibrated against clinical time courses from eight acute HBV studies. Perturbation and local sensitivity analyses identified CD8⁺ T cell activity, B cell responses, dendritic cell priming, and viral replication processes (NTCP-mediated entry, cccDNA formation) as dominant drivers of biomarker dynamics. A virtual population generated by varying the most influential parameters successfully reproduced the interindividual variability observed in clinical HBV DNA, HBsAg, and ALT trajectories. The model provides a mechanistic foundation for future extensions to chronic HBV infection and for evaluating antiviral and immunomodulatory combination therapies.

Scientific Context & Motivation

Chronic HBV infection affects over 250 million people worldwide and causes significant morbidity and mortality. The mechanisms determining whether acute infection resolves or becomes chronic are incompletely understood, partly due to the complexity of host-virus interactions. Previous mathematical models of HBV simplified immune processes, lacking key features such as NTCP-mediated viral entry, differentiation of CD4⁺ T cell subsets (Th1, Th2, TFH, Treg),

cytokine-mediated regulation (IL-10, TGF- β), and distinction between cccDNA- and integrated DNA-harboring hepatocytes. This model addresses these gaps by integrating detailed virological and immunological mechanisms, providing a platform to quantitatively explore infection outcomes and therapeutic interventions.

✂ Methodological Snapshot

The QSP model comprises 98 species and 319 parameters, organized into five interconnected modules: viral life cycle, innate immunity, adaptive immunity, cytokine signaling, and viral clearance. The model operates across liver, plasma, and lymph compartments. Parameterization used published literature and databases (CYTOCON DB, FIVE DB, Immune Response Template), with 45 parameters manually calibrated against clinical time-course data from eight acute HBV studies. Model evaluation included qualitative visual comparison, quantitative kinetic metric analysis (peak magnitude/timing, rising/clearance slopes), perturbation analyses (immune pathway knockouts), local sensitivity analysis (derivative-based), and virtual population generation using a Markov chain approach on the top 20 most sensitive parameters per biomarker.

☒ Structural Model Breakdown

The model includes 98 species and 319 parameters across three compartments. Hepatocytes are divided into uninfected, cccDNA-containing infected, and iDNA-containing infected subpopulations. Viral life cycle includes NTCP receptor binding, internalization, cccDNA formation, transcription, translation, assembly, and release. Innate immunity includes NK cells (naïve and activated), Kupffer cells (resting and activated), and dendritic cells (mDC and pDC, resting and activated). Adaptive immunity includes CD4⁺ T cells (naïve, Th1, Th2, TFH, Treg), CD8⁺ T cells (naïve, effector, memory), and B cells (naïve, activated, plasmablasts, plasma cells). Cytokine module includes 10 cytokines with production, degradation, and receptor binding. Antibody module includes anti-HBs, anti-HBe, anti-HBc. Viral clearance mechanisms: cytokine-mediated non-cytolytic suppression, CD8⁺ and NK cell-mediated cytotoxicity, antibody neutralization and immune complex formation.

Detailed Methodological Analysis

Modeling Approach

Mechanistic QSP model built in MATLAB SimBiology R2024a. Five interconnected modules: viral life cycle (NTCP entry, cccDNA/iDNA hepatocytes, antigen production), innate immunity (NK cells, Kupffer cells, dendritic cells), adaptive immunity (CD4⁺ subsets, CD8⁺ T cells, B cells, antibodies), cytokine signaling (IFN- γ , IFN- α , TNF- α , IL-2, IL-4, IL-6, IL-10, IL-12, IL-21, TGF- β), and viral clearance (cytokine-mediated suppression, cytotoxicity, antibody neutralization). Three compartments: liver, plasma, lymph.

Data Sources

Clinical biomarker data from eight acute HBV studies (Webster et al., Whalley et al., Chulanov et al., Fiscaro et al., Dunn et al., Okamoto et al., Kuhns et al., Panduro et al.) including longitudinal HBV DNA, HBsAg, HBcAg, HBeAg, anti-HBs, anti-HBc, ALT, immune cell counts, and cytokines. Data extracted via digitization (Graph Grabber) and standardized to common time reference. Parameter estimates from peer-reviewed literature and curated databases (CYTOCON DB, fIVE DB, Immune Response Template).

Estimation Methods

Manual calibration of 45 parameters due to model complexity and limited acute HBV datasets preventing automated optimization convergence. Parameters adjusted to match clinical time-course data for HBV DNA, HBsAg, and ALT.

Model Evaluation

Qualitative visual comparison of simulated vs. observed trajectories. Quantitative comparison using kinetic metrics: peak value and time, rising slope (log10/day for HBV DNA and HBsAg, linear for ALT), single-phase clearance slope, and biphasic clearance slope for HBV DNA (with breakpoint selection via BIC). Virtual population generation with Markov chain sampling of top 20 sensitive parameters ($\sigma = 0.02 \times \text{parameter value}$).

Covariate Analysis

Not applicable (no traditional covariate analysis). Local sensitivity analysis identified influential parameters for HBV DNA, SVP (HBsAg proxy), and ALT. Top 20 parameters per biomarker used for virtual population generation.

Statistical Rigor Assessment

The model evaluation uses both qualitative and quantitative approaches. Quantitative metrics (peak, slopes) are compared between simulated and observed data using boxplots and jittered points, with simulated values falling within interquartile ranges. Biphasic clearance assessment uses BIC for breakpoint selection. Virtual population generation uses a Markov chain approach, but no formal statistical inference (e.g., confidence intervals, hypothesis tests) is reported. Structural and practical identifiability analyses were not performed due to model complexity. The manual calibration approach limits statistical rigor, but the authors acknowledge this and provide transparent reporting of parameter sources and calibration procedures.

Key Findings

The model accurately reproduces hallmark features of acute HBV infection: exponential viral rise, peak viremia, biphasic viral clearance, transient ALT elevation (peak ~10.8% hepatocyte loss), and resolution with antibody persistence. Perturbation analyses show that CD8+ T cell or mDC knockout leads to uncontrolled viral replication; B cell or TFH knockout results in partial clearance with chronic plateau; enhanced Treg differentiation produces low-level chronic infection with normal ALT. Local sensitivity analysis identifies 23 key parameters: viral replication

(cccDNA formation, NTCP entry, HBV degradation), B cell dynamics (plasmablast proliferation/death, IL-4/IL-21 signaling), T cell/antigen presentation (CD8+ differentiation, mDC activation), and hepatocyte turnover. Virtual population captures interindividual variability in biomarker peak magnitude and timing.

🔗 Clinical & Regulatory Implications

The model provides a mechanistic framework for understanding acute HBV infection dynamics and identifying key drivers of viral clearance versus chronicity. Clinically, the model suggests that interventions enhancing CD8+ T cell activity and B cell responses, while modulating Treg activity, could promote viral clearance. The virtual population approach can inform clinical trial design by predicting expected biomarker trajectories and variability. For drug development, the model can evaluate candidate therapies targeting NTCP entry, cccDNA formation, or immune checkpoints. However, direct clinical recommendations require extension to chronic HBV and validation against treatment data.

Strengths & Limitations

Strengths

- Most comprehensive QSP model of acute HBV to date, integrating detailed virological and immunological mechanisms absent in previous models
- Successful calibration against multiple clinical datasets with both qualitative and quantitative evaluation
- Perturbation and sensitivity analyses systematically identify key drivers of infection dynamics
- Virtual population generation captures interindividual variability, providing foundation for chronic HBV extension
- Model designed for future therapeutic evaluation (antiviral, immunomodulatory, combination therapies)

Limitations (Acknowledged by Authors)

- Manual calibration of 45 parameters due to model complexity and limited data preventing automated optimization
- Structural and practical identifiability analyses not performed due to computational infeasibility
- Global sensitivity analysis numerically unstable, results not shown
- Time adjustments for clinical data required due to uncertain infection onset
- Perturbation analyses represent drastic single-pathway manipulations, not realistic partial impairments

Limitations (Expert Review)

- No formal uncertainty quantification or confidence intervals for parameter estimates
- Virtual population generation uses only top 20 parameters per biomarker, potentially missing interactions

- Model validation limited to acute resolving infection; chronic extension requires reparameterization
- Data from only eight studies with small sample sizes (total ~40 patients), limiting generalizability
- No external validation dataset used; all data used for calibration
- Cytokine dynamics (e.g., IFN- α , ...) show only transient spikes, consistent with ‘stealth virus’ concept but may underestimate innate responses

Generalizability

The model is specifically calibrated for acute HBV infection in immunocompetent adults and may not generalize to perinatal infection, different HBV genotypes, or immunocompromised populations. The virtual population captures variability within acute resolving infections but has not been validated for chronic HBV. Extension to chronic infection will require reparameterization of immune parameters. The model structure is adaptable to other hepatotropic viruses with similar biology.



Figures & Tables

- **Figure 1:** Overview of the acute HBV QSP model structure showing interactions between uninfected/infected hepatocytes, immune cells, and molecular mediators across liver, plasma, and lymph compartments.
 - *Significance:* Provides a visual summary of the model’s scope and interconnected modules, essential for understanding the framework.
- **Figure 2:** Comparison of model predictions (solid lines) with clinical data (points) for 12 plasma biomarkers: HBV DNA, ALT, HBsAg, HBcAg, HBeAg, anti-HBs, anti-HBc, Th1 cells, CD8+ T cells, NK cells, IFN- α , and IL-10.
 - *Significance:* Demonstrates the model’s ability to reproduce the characteristic time course of acute HBV infection across multiple biomarkers, validating the mechanistic assumptions.
- **Figure 3:** Quantitative comparison of kinetic metrics (peak value, peak time, rising slope, clearance slope) for HBV DNA, ALT, and HBsAg between clinical data (boxplots) and simulations (dashed lines with diamonds).
 - *Significance:* Provides a rigorous quantitative assessment showing simulated metrics fall within interquartile ranges of clinical observations, supporting model credibility.
- **Figure 4:** Impact of perturbation scenarios (immune pathway knockouts and enhanced Treg differentiation) on simulated HBV DNA, ALT, and HBsAg time courses.
 - *Significance:* Identifies which immune components are essential for viral clearance and reveals that different perturbations produce phenotypes resembling clinical chronic HBV phases.
- **Figure 5:** Local sensitivity analysis results showing normalized integrated sensitivity values for the top 20 parameters influencing HBV DNA, SVP (HBsAg proxy), and ALT.
 - *Significance:* Quantifies the relative importance of viral replication, B cell, and T cell parameters, guiding future experimental and modeling efforts.

- **Figure 6:** Virtual population simulations (gray lines) overlaid on clinical data (colored points) for ALT, HBsAg, and HBV DNA, showing the model's ability to capture interindividual variability.
 - *Significance:* Demonstrates that variability in key parameters can reproduce the range of clinical trajectories, supporting the model's utility for population-level predictions.
 - **Table 1:** Summary of eight acute HBV studies used for model calibration, including patient numbers, biomarkers measured, and key characteristics.
 - *Significance:* Documents the clinical data sources and their specific contributions to model calibration, essential for reproducibility and assessing data coverage.
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Code & Reproducibility Assessment

The model was developed in MATLAB SimBiology R2024a. No code or model files are publicly provided. Parameter values and initial conditions are listed in Appendix S3 (supplementary materials). The authors used commercial databases (CYTOCON DB, fIVE DB, Immune Response Template) which may not be publicly accessible.

Supplementary Materials

Supplementary materials include: Appendix S1 (Supplementary Material 1) with detailed model structure description, generic mathematical expressions (Table S1), steady-state validation (Table S2), time adjustment details (Table S3), and additional figures (Figures S1-S2). Appendix S2 (Supplementary Material 2) contains all model equations. Appendix S3 lists all parameter values, initial conditions, and sources. Appendix S4 includes Table S5 with summary statistics of accepted virtual population parameter distributions.

Future Directions

The authors plan to extend the model to chronic HBV infection by reparameterizing immune parameters to reflect immune dysfunction (T cell exhaustion, Treg-mediated suppression). Future work will represent key chronic phases: immune tolerance, immune clearance, inactive carrier, and occult infection. The model can be used to evaluate antiviral agents (e.g., entry inhibitors, cccDNA silencers) and immunomodulatory therapies (e.g., TLR agonists, checkpoint inhibitors, therapeutic vaccines). Combination therapy simulations could assess synergy and probability of functional cure. Experimental studies to constrain key parameters (e.g., mDC activation sensitivity, B cell maturation rates) are needed.

Expert Commentary

This model represents a significant step forward in HBV QSP modeling by incorporating detailed immune mechanisms that previous models omitted. The perturbation analysis is particularly insightful, showing that different immune knockouts produce phenotypes resembling clinical chronic HBV phases (immune-active, immune-tolerant, inactive carrier). This suggests the model has biological validity beyond acute infection. However, the manual calibration of 45 param-

ters is a concern for reproducibility and uncertainty quantification. The lack of identifiability analysis is understandable given model complexity, but future work should attempt profile likelihood or MCMC approaches on reduced models. The virtual population generation using only 20 parameters is a pragmatic approach but may miss important parameter interactions. For practicing pharmacometricians, this model demonstrates the value of mechanistic modeling for understanding complex immunovirological dynamics, but the path to regulatory application will require more rigorous statistical evaluation and validation in chronic HBV populations.

Bottom Line

This QSP model provides the most detailed mechanistic framework to date for acute HBV infection, successfully reproducing clinical biomarker dynamics and identifying CD8+ T cells, B cells, and viral replication processes as key determinants of infection outcome. The model is a valuable platform for hypothesis generation and will support future extensions to chronic HBV and evaluation of curative therapies. Pharmacometricians should note the manual calibration approach and lack of formal identifiability analysis as limitations, but the comprehensive biological detail and successful reproduction of clinical data make this a significant contribution to HBV modeling.

☒ Figures

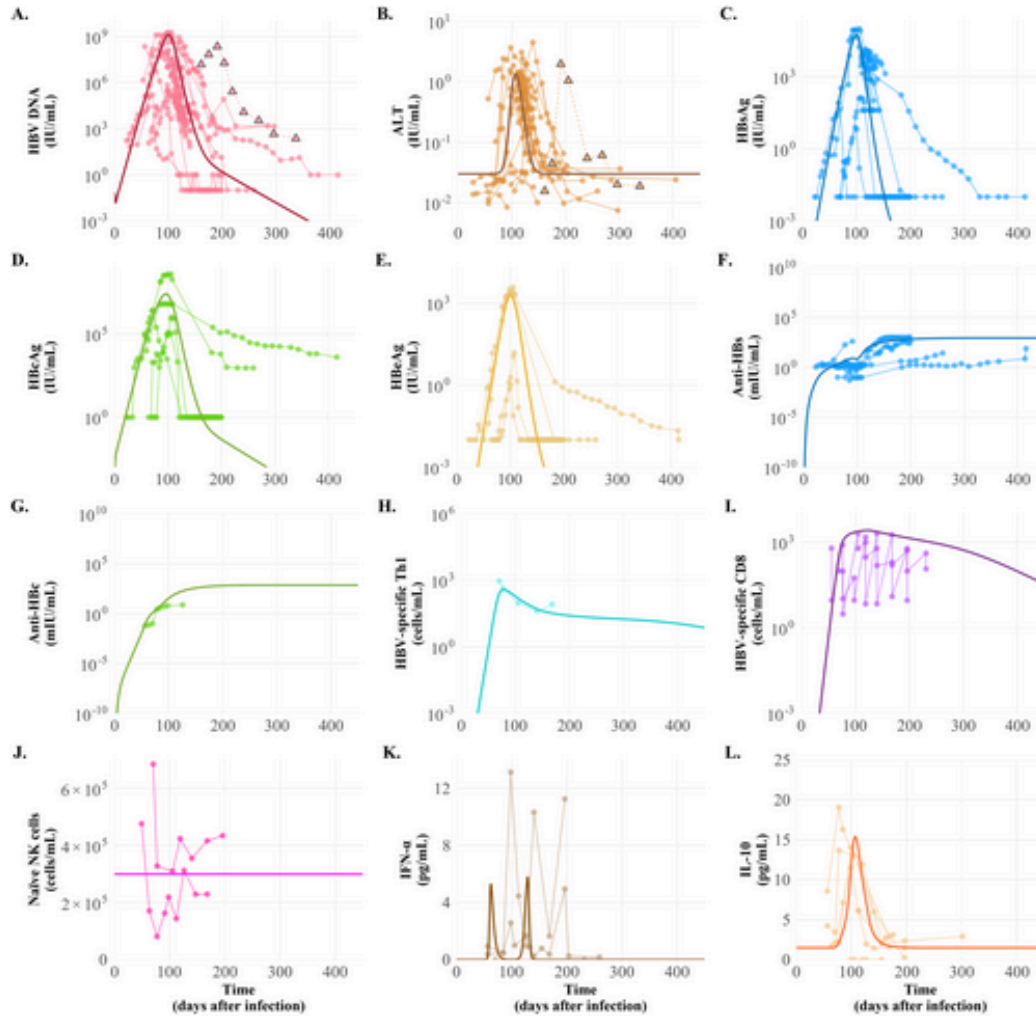


Figure 2: Comparison of model predictions and clinical data for plasma biomarkers. Simulated trajectories (solid lines) of key plasma biomarkers are shown following model

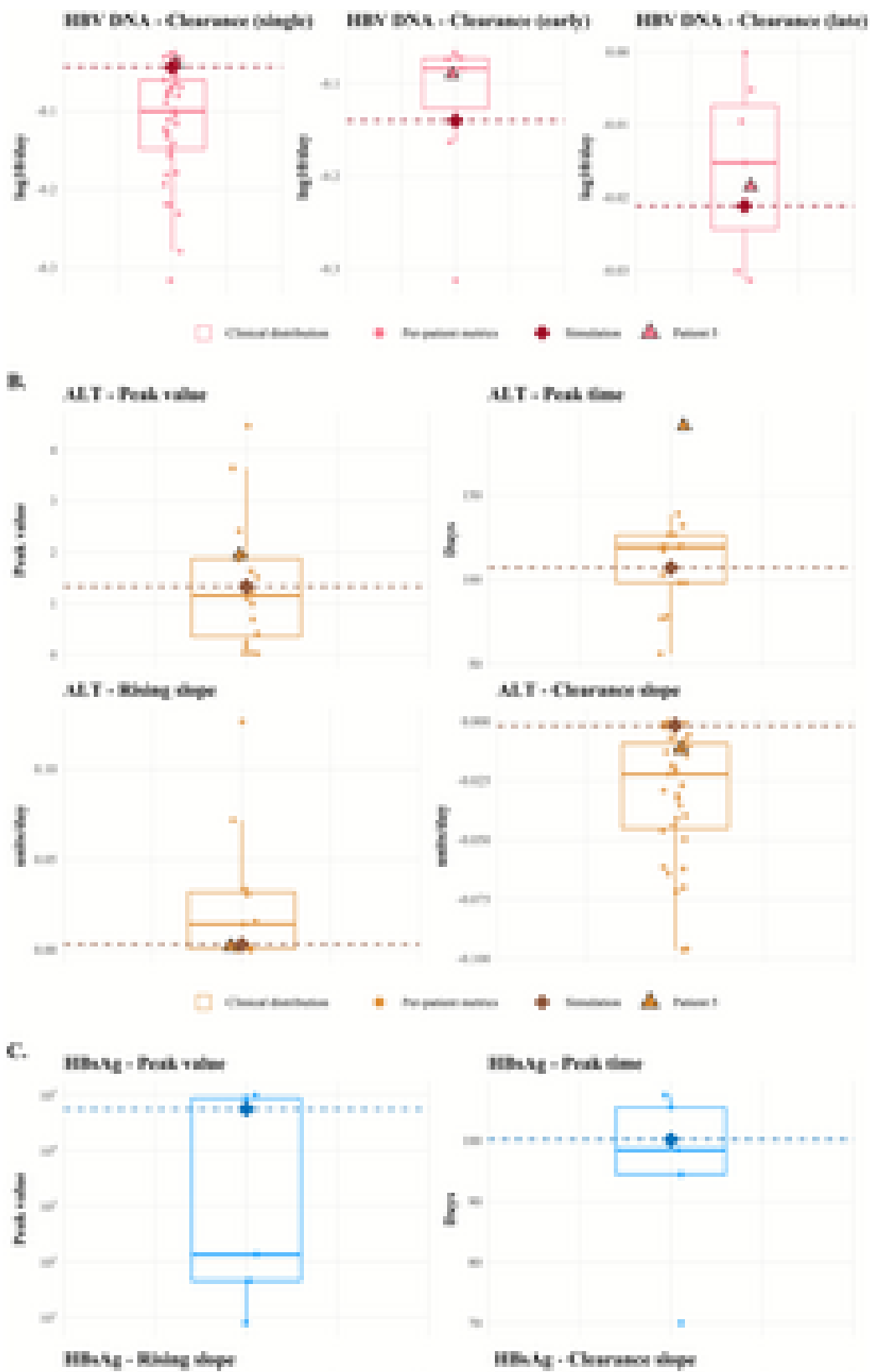


Figure 3: Quantitative comparison of simulated and clinical biomarker kinetics. Panels A–C show kinetic metrics for HBV DNA (A), ALT_0 (B), and HBsAg (C). Clinical data are

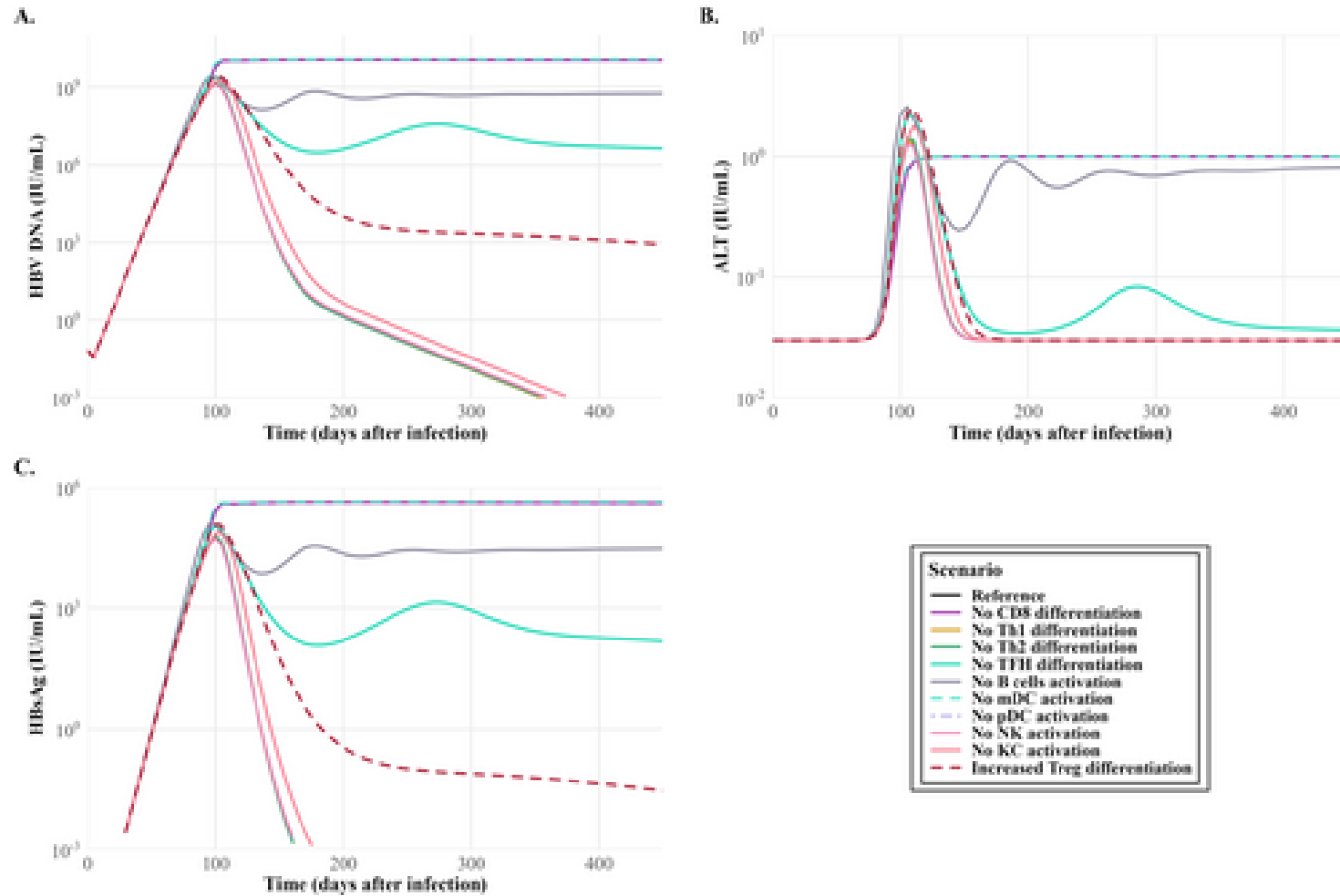


Figure 4: Impact of perturbation scenarios on model-predicted plasma biomarkers. Simulated time courses of HBV DNA (A), ALT (B), and HBsAg (C) concentrations in plasma und

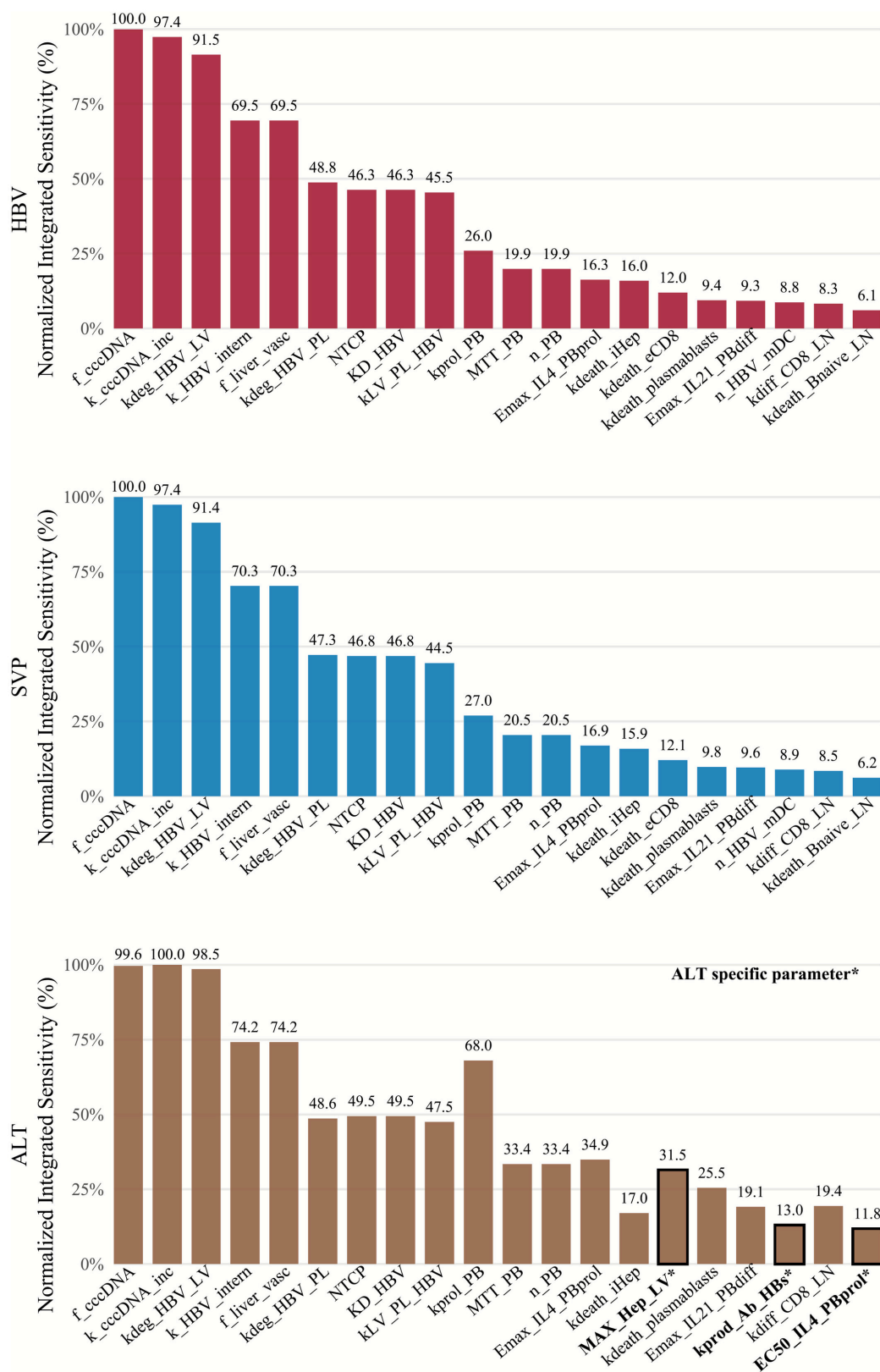


Figure 5: LSA results for key plasma biomarkers. Normalized integrated sensitivity values (expressed as percentages) are shown for HBV DNA, SVP, and ALT, highlighting the

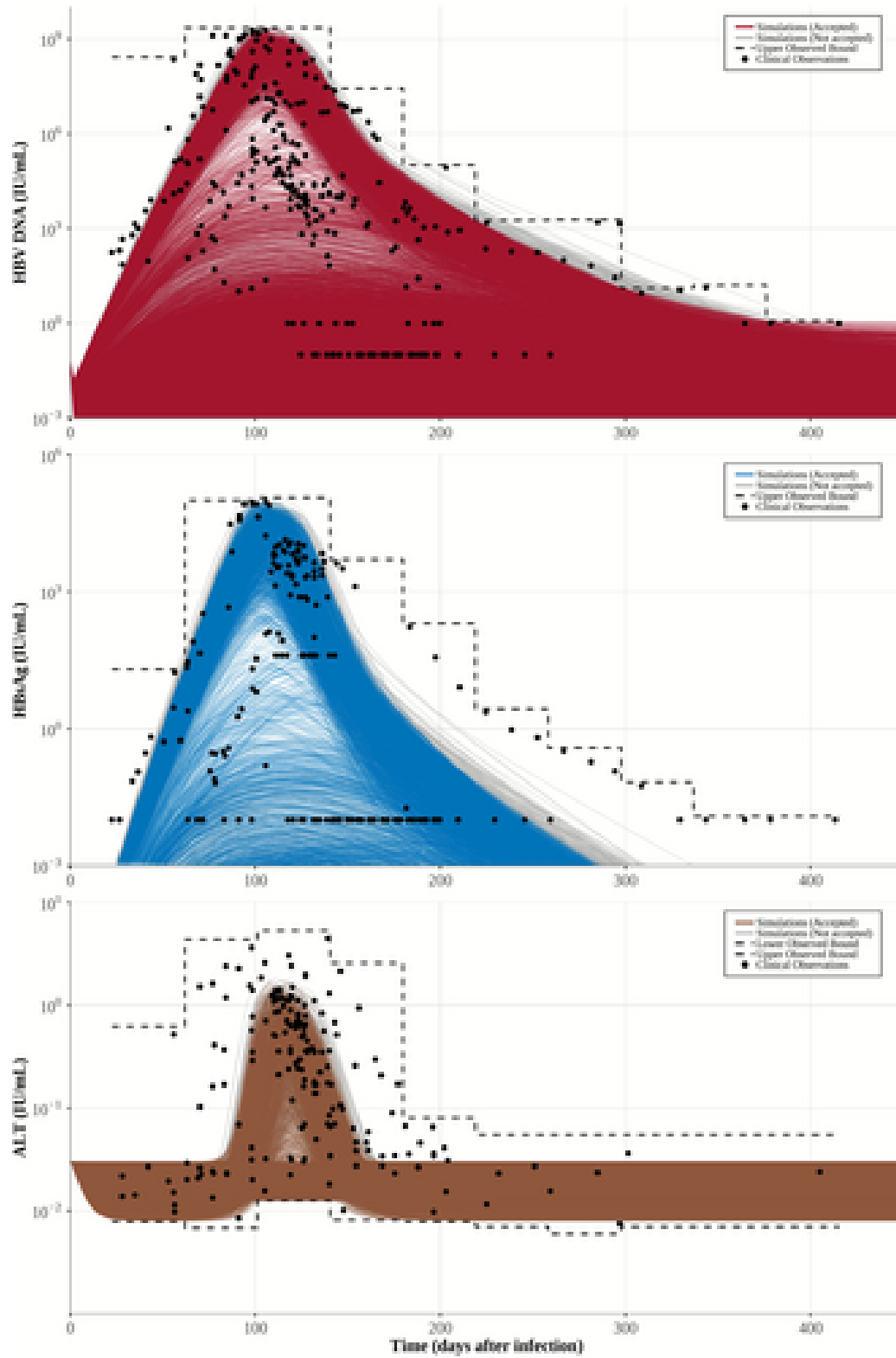


Figure 6: Virtual population calibration to clinical biomarker data. The generated virtual population was calibrated to capture the clinical variability observed in ALT, H